

Backtrace

```
mari@DESKTOP-0K07MJA:~/xv6-labs-2025$ addr2line -e kernel/kernel 80001e3e
/home/mari@DESKTOP-0K07MJA:~/xv6-labs-2025/kernel/sysproc.c:75
mari@DESKTOP-0K07MJA:~/xv6-labs-2025$ addr2line -e kernel/kernel 80001d18:
/home/mari@DESKTOP-0K07MJA:~/xv6-labs-2025/kernel/syscall.c:141 (discriminator 1)
mari@DESKTOP-0K07MJA:~/xv6-labs-2025$ addr2line -e kernel/kernel 80001a9c
/home/mari@DESKTOP-0K07MJA:~/xv6-labs-2025/kernel/trap.c:80
mari@DESKTOP-0K07MJA:~/xv6-labs-2025$ addr2line -e kernel/kernel 80001d18
/home/mari@DESKTOP-0K07MJA:~/xv6-labs-2025/kernel/syscall.c:141 (discriminator 1)
mari@DESKTOP-0K07MJA:~/xv6-labs-2025$
```

```
mari@DESKTOP-0K07MJA: x + v
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attribute
AB_TRAPS -MD -mmodel=medany -ffreestanding -fno-common -nos
-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builti
rchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc
n-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-
bttest.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld
printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_bttest > user/bttest.as
riscv64-unknown-elf-objdump -t user/_bttest | sed '1,/SYMBOL
mkfs/mkfs fs.img README user/_cat user/_echo user/_forktest
dir user/_rm user/_sh user/_stressfs user/_usertests user/_g
r/_dorphan user/_call user/_bttest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitma
balloc: first 992 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -dev

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ bttest
backtrace:
80001e3e
80001d18
80001a9c
$ |
```

Due to difference in printing output the make grade for backtrace fails..cant get the output format to look the same as the one asked in the lab

Alarm

```
mariu@DESKTOP-0K07MJA: x + v
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virti

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ alarmtest
test0 start
.....alarm!
test0 passed
test1 start
..alarm!
..alarm!
..alarm!
..alarm!
..alarm!
..alarm!
..alarm!
..alarm!
..alarm!
..alarm!
test1 passed
test2 start
.....alarm!
test2 passed
test3 start
test3 passed
$ usertests -q
usertests starting
```

```
mariu@DESKTOP-0K07MJA: x + v
usertrap(): unexpected scause 0xf pid=6631
sepc=0x4442 stval=0x33012000
usertrap(): unexpected scause 0xf pid=6632
sepc=0x4442 stval=0x34012000
usertrap(): unexpected scause 0xf pid=6633
sepc=0x4442 stval=0x35012000
usertrap(): unexpected scause 0xf pid=6634
sepc=0x4442 stval=0x36012000
usertrap(): unexpected scause 0xf pid=6635
sepc=0x4442 stval=0x37012000
usertrap(): unexpected scause 0xf pid=6636
sepc=0x4442 stval=0x38012000
usertrap(): unexpected scause 0xf pid=6637
sepc=0x4442 stval=0x39012000
usertrap(): unexpected scause 0xf pid=6638
sepc=0x4442 stval=0x3a012000
usertrap(): unexpected scause 0xf pid=6639
sepc=0x4442 stval=0x3b012000
usertrap(): unexpected scause 0xf pid=6640
sepc=0x4442 stval=0x3c012000
usertrap(): unexpected scause 0xf pid=6641
sepc=0x4442 stval=0x3d012000
usertrap(): unexpected scause 0xf pid=6642
sepc=0x4442 stval=0x3e012000
usertrap(): unexpected scause 0xf pid=6643
sepc=0x4442 stval=0x3f012000
OK
test lazy_copy: OK
ALL TESTS PASSED
$ |
```

Make Grade

```
mariu@DESKTOP-0K07MJA: × + v
make[1]: Leaving directory '/home/mariu/xv6-labs-2025'
== Test answers-traps.txt ==
answers-traps.txt: OK
== Test backtrace test ==
$ make qemu-gdb
backtrace test: FAIL (5.4s)
    got:
        0
    expected:
        3
    QEMU output saved to xv6.out.backtracetest
== Test running alarmtest ==
$ make qemu-gdb
(6.2s)
== Test alarmtest: test0 ==
alarmtest: test0: OK
== Test alarmtest: test1 ==
alarmtest: test1: OK
== Test alarmtest: test2 ==
alarmtest: test2: OK
== Test alarmtest: test3 ==
alarmtest: test3: OK
== Test usertests ==
$ make qemu-gdb
usertests: OK (117.5s)
== Test time ==
time: OK
Score: 85/95
make: *** [Makefile:370: grade] Error 1
mariu@DESKTOP-0K07MJA:~/xv6-labs-2025$
```